

Alejandro Solís Mercado

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SUMMARY

Mechanical Engineering graduate with background in engineering design, simulation, and programming. Led multidisciplinary engineering teams across 5 international aerospace, robotics, and automotive student competitions. I've used SolidWorks CAD/CAM, DFM, FEM/CFD Simulation, GD&T, to produce functional prototypes. Proficient in making 3D CAD models.

EDUCATION

Monterrey Institute of Technology, Toluca Campus

August 2022 – June 2026

B.Sc. Mechanical Engineering – Grade Average: 94/100 – Academic Excellence Scholarship

Awards: National Award for Excellence EGEL - CENEVAL (2026), Excellence in Integral Formation (2026)

Universiteit Twente, Netherlands – Study Abroad

August 2025 – January 2026

Minor in Aerospace Management and Engineering

ACADEMIC EXPERIENCE

DroneTeam Universiteit Twente

August 2025 – January 2026

Aerodynamics and Design Engineer

Enschede, Netherlands

- Collaborated with a 26-person team of international students to conceptualize and build a 10 kg VTOL tricopter.
- Communicated in technical English to establish the aerodynamic design methodology and performance analyses.
- Designed the outer wing section and aileron mechanism in SolidWorks for LW-PLA 3D-printing.
- Optimized aerodynamic performance using SW Flow, MATLAB, Flow5, achieving 41% higher peak airfoil efficiency.

General Motors

August 2024 – October 2024

Jr. Consultant – University Coursework

- Won first place in GM's "Morphing Pocket" Challenge with a SolidWorks-designed 3D-printed prototype of a smart map pocket for a car's door trim.
- Developed a concentric-tube heat exchanger study comparing analytical (Python), CFD (SolidWorks Flow), and experimental data; cutting computation time from 40 min to under 1 s while maintaining 7% error, enabling rapid parametric studies of tube dimensions and materials.

EXTRACURRICULAR PROJECTS

Student Team Xinantec Aerodesign

August 2023 – July 2026

President & Lead Engineer

- Developed engineering analysis using XFLR5, SolidWorks, Ansys CFD, and MATLAB for design optimization of various electric aircraft with 2.5-3.5 m wingspan, 5-10 kg weight, and 750 W powertrain for single-lap payload missions.
- Achieved 3rd place among 20 universities in SAE Aerodesign MX 2023's flight rounds.
- Coordinated sponsorships with Altair and Ansys, accessing CFD tools that enabled higher-fidelity simulations.

Student Team Borregos Toluca Baja Racing

February 2025 – June 2025

Mechanical Designer

- Designed, manufactured and analytically validated gearbox drivetrain shafts for 110 Nm torque in a Baja SAE vehicle using SolidWorks and MATLAB, performing analysis on AISI 1020 steel to achieve 2.7+ safety factor under bending and torsion.

FRC Student Team TecBot 3158

August 2018 – June 2022

Design and Programming Leader

- Led mechanical design in SolidWorks and programming of Java control systems for a 40-member FRC team.
- Designed an aluminum and polycarbonate pneumatic-actuated ball-intake mechanism with MDF prototyping and PID-tuned automation that improved consistency, achieving 4th best autonomous scoring in competition.

SKILLS AND CERTIFICATIONS

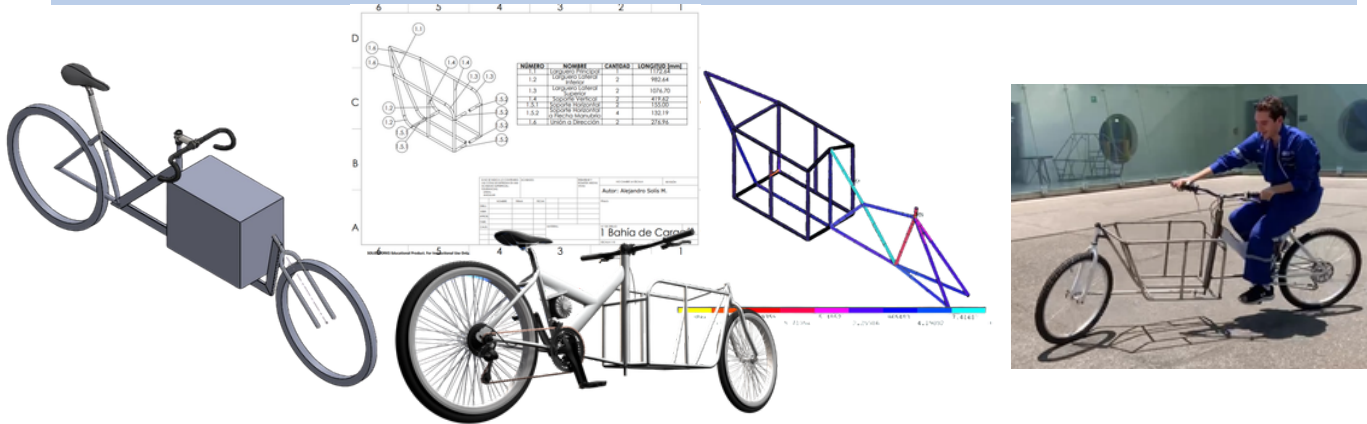
- **Certifications:** Certified SolidWorks Professional (*Dassault Systèmes*), Scrum Fundamentals Certified (*SCRUMstudy*).
- **Skills:** CAD Design (SolidWorks, Fusion360, NX), Structural/Aerodynamic/Thermodynamic Simulation (FEM+CFD in ANSYS, Altair, AutoDesk, SW), Programming (Java, Python, MATLAB, C++), Office (Excel), LLM-Assisted Scripting.
- **Languages:** Spanish (Native), English (Advanced, C2), French (Basic), German (Basic).

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Electric Cargo Bike - University Coursework



What?

- Convert an existing mountain bike into an electric cargo bicycle with capacity to transport a driver and a 70 kg, 60 L payload with 1.5+ FoS.
- Evaluate design with FEA.
- Implement **DFM**.

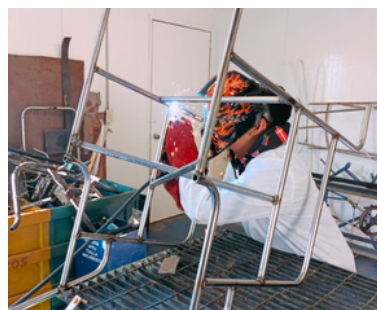
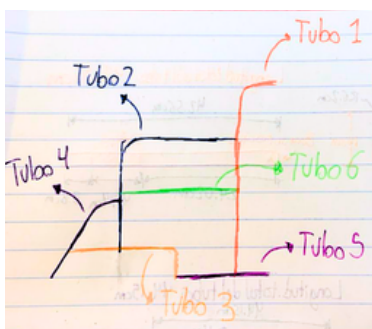
How?

- Modeled elements and **brackets** using **SolidWorks Weldments**
- Created **drawings** and **BoM**
- Evaluated **FoS** with **FEA** in **Ansys Mechanical APDL**
- **Soldered** electrical components
- Cut, bent, and **MIG welded** frame

Results

- The finalized prototype was built for under \$15K MXN.
- Electric motor was successfully user-operated
- It was fully functional and was able to carry more weight than designed

Wheelchair - DIF Charity Coursework



What?

- Create a low-cost wheelchair as charity coursework for the local social assistance agency
- Design around existing wheelbase.
- Implement **DFM**.

How?

- Modeled tubular elements using **SolidWorks Weldments**
- Produced **2D drawings** for manufacture
- Cut, bent, and **MIG welded** the frame
- Designed accessories in **SolidWorks**
- **3D-Printed** accessories

Results

- The finalized product was built for under \$12K MXN
- It was successful in carrying a human and is still functional 3 years later

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Gearbox Axles - Team Borregos Toluca Baja Racing



What?

- Design transmission axles for an existing Baja SAE gearbox casing (14 HP, 2500 RPM)
- Validate the design analytically
- Manufacture axles

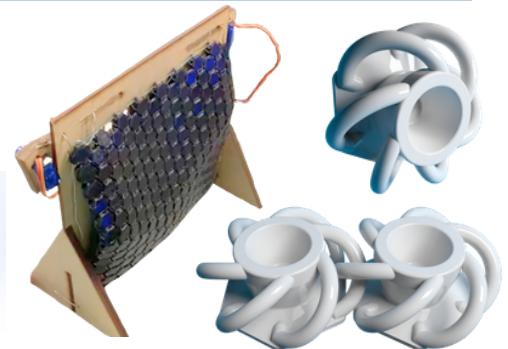
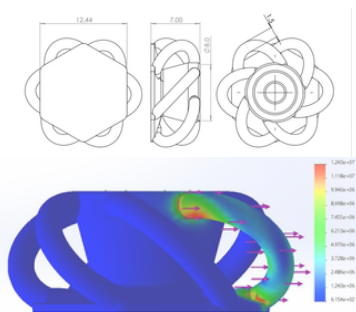
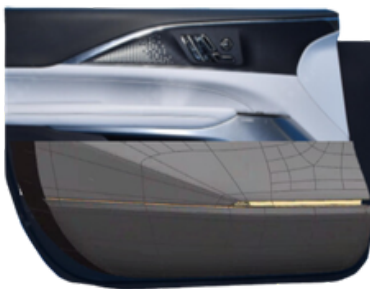
How?

- **Measured** existing casing with height gauge
- **Modeled** 3D shaft geometry and **2D drawings** using **SolidWorks**
- Programmed stress analysis scripts in **MATLAB**
- Manufactured axles in **lathe**

Results

- The design was validated to handle peak output 110Nm with 2.7+ safety factor under combined loads
- The gearbox was assembled with the manufactured components

Morphing Pocket - General Motors Challenge



What?

- GM's 'Map Pocket Challenge' was to use *Design Thinking* Methodology to create an invisible, aesthetic, and functional map pocket for automotive door trim.

How?

- Our solution was to use servo-actuated open-source, **3D-printed** interlocking chainmail for a Smart 'Morphing Pocket'
- Used **SolidWorks Simulation** to perform **FEA** on link elements to guarantee function
- Constructed a working prototype

Results

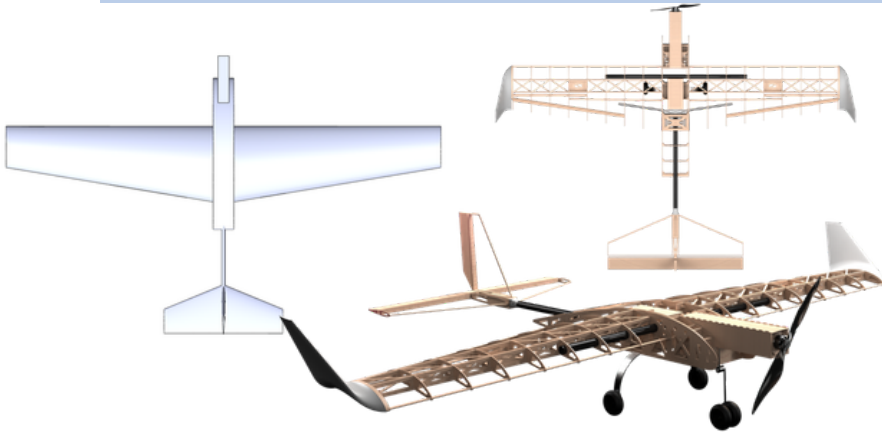
- Our team was awarded 1st place by General Motors engineers at Toluca Regional Engineering Center

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Coquetec - Team Xinantec Aerodesign



What?

- Design, build, and fly an electric cargo aircraft with a 750 W power limit for a single-lap mission
- Handle solid and liquid payload

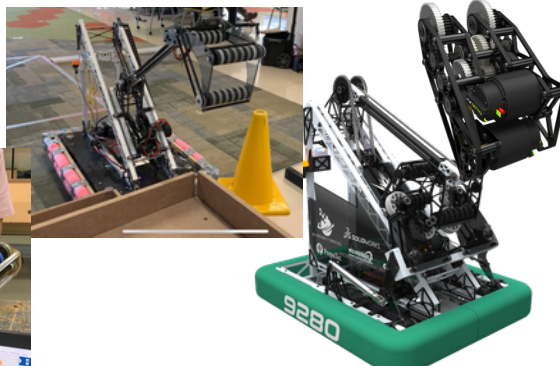
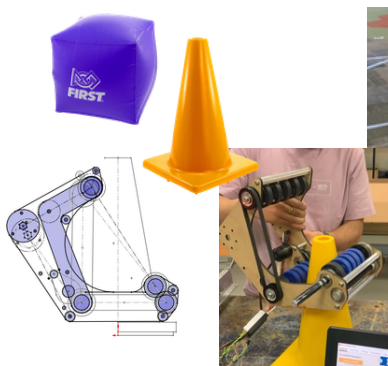
How was I involved?

- Optimized aerodynamics with **Solidworks Flow, XFLR5** and **MATLAB**
- Modeled with **SolidWorks**
- **Laser-cut, 3D-Printed** and assembled components

Results

- The 4.5kg plane completed a lap with a 3kg payload
- 3rd best flight performance amidst top aeronautical universities despite our university not having that major

Thundercrash - Team High Altitude Robotics



What?

- Create a robot for the First Robotics Competition in a 40 person team.
- Pick and place cones and inflatable cubes in a field with 5 other robots
- Win by scoring the most cones and cubes

How was I involved?

- MDF Laser-cut prototyping
- Detailed design of **gearboxes, pneumatic systems, pulley** and **chain** systems in **SolidWorks**
- **Programmed PID-tuned Java** routines for autonomous and manual control

Results

- The robot competed in Mexico and US. It was ranked 13th/85 Mexican teams
- My autonomous routine was 4th/35 best in competition